Python String Methods — DSA Cheat Sheet

A practical reference for LeetCode, coding interviews, and everyday Python string manipulation.

1. Creating, Indexing & Slicing

Method / Operation	Syntax / Example
Create	`s = "hello"`
Index	`s[0] # 'h' | s[-1] # 'o'`
Slice	`s[1:4] # 'ell' | s[:3] | s[2:]`
Reverse	`s[::-1] # 'olleh'`
Length	`len(s)`

2. Case Conversion

Method / Operation	Syntax / Example
lower()	`s.lower() # "hello world"`
upper()	`s.upper() # "HELLO WORLD"`
capitalize()	`s.capitalize() # first character uppercase`
title()	`s.title() # "Hello World"`
swapcase()	`s.swapcase() # swap upper/lower case`

3. Whitespace

Method / Operation	Syntax / Example
strip()	`s.strip() # remove whitespace from both ends`
lstrip()	`s.lstrip() # remove from left`
rstrip()	`s.rstrip() # remove from right`

4. Searching

Method / Operation	Syntax / Example
find()	`s.find("x") # index of first occurrence, -1 if absent`
rfind()	`s.rfind("x") # search from right`
index()	`s.index("x") # like find(), but raises ValueError if absent`
in	`"abc" in s # True/False`
not in	`"abc" not in s`

5. Counting & Replacing

Method / Operation	Syntax / Example
count()	`s.count("a") # number of occurrences`
replace()	`s.replace("old", "new")`
Limited replace	`s.replace("a", "b", 1) # replace at most 1 occurrence`

6. Split & Join

Method / Operation	Syntax / Example
split()	`s.split() # split on whitespace`
split(separator)	`s.split(",") # split using a delimiter`
join()	`" ".join(words) # list -> string`
Remove spaces	`"".join(s.split())`

7. Character Checks

Method / Operation	Syntax / Example
isalpha()	`"abc".isalpha() # letters only`
isdigit()	`"123".isdigit() # digits only`
isalnum()	`"abc123".isalnum() # letters or digits`
isspace()	`" ".isspace() # whitespace only`
islower()	`"abc".islower()`
isupper()	`"ABC".isupper()`

8. Prefix / Suffix

Method / Operation	Syntax / Example
startswith()	`s.startswith("hello")`
endswith()	`s.endswith("world")`
Multiple choices	`s.startswith(("hi", "hello"))`

9. Conversion & Built-ins

Method / Operation	Syntax / Example
String -> list	`list("hello") # ["h", "e", "l", "l", "o"]`
List -> string	`"" .join(["h", "i"]) # "hi"`
Sort characters	`sorted("dcba") # ["a", "b", "c", "d"]`
Sorted string	`"".join(sorted("dcba")) # "abcd"`
Minimum / maximum	`min(s), max(s)`

10. Useful Formatting

Method / Operation	Syntax / Example
f-string	`name = "Cheery"; age = 22`
	`f"My name is {name} and I am {age}"`

DSA Patterns to Memorize

Palindrome / two-pointer

```
left = 0
right = len(s) - 1
while left < right:
if s[left] != s[right]:
return False
left += 1
right -= 1
```

Frequency dictionary

```
freq = {}
for ch in s:
freq[ch] = freq.get(ch, 0) + 1
```

Seen set

```
seen = set()
for ch in s:
if ch in seen:
# duplicate
...
seen.add(ch)
```

Normalize before comparison

```
s = s.lower().strip()
```

Remove non-alphanumeric characters

```
clean = "".join(ch for ch in s if ch.isalnum())
```

Reverse a string

```
reversed_s = s[::-1]
```

Anagram check

```
sorted(s1) == sorted(s2)
```

Important: Strings are immutable

Methods such as **upper()**, **lower()**, **replace()**, and **strip()** return a new string. They do not modify the original string. Use **s = s.upper()** when you want to keep the result.

Quick Priority List for LeetCode

Must know: len(), indexing, slicing, lower(), upper(), strip(), split(), join(), find(), count(), replace(), in, startswith(), endswith(), isalpha(), isdigit(), isalnum(), list(), sorted(), and [::-1].